



Agronomic Responses and Solute Leaching Under Fertigation With Synthetic Wastewater in Two Nutrient-poor Sandy Soils

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Received: 23 April 2026 / Accepted: 8 September 2026
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Abstract

In this study, the agronomic response of radish and the leaching behavior of nutrients, salts and trace elements were investigated under fertigation with synthetic waters reproducing treated domestic wastewater, industrial effluent and landfill leachate. The water compositions were derived from a systematic review of 75 wastewater-fertigation studies and were evaluated in two nutrient-poor sandy soils through batch tests and a 40-day laboratory pot experiment. Mixed-effects models identified significant Soil × Water × Time interactions for 13 of the 14 drainage-water variables. Nitrate, sodium, chloride and potassium showed high mobility, whereas trace-element concentrations in drainage water remained low. For biomass, Soil and Water had significant main effects on all variables, whereas no Soil × Water interaction was detected. The nutrient-containing waters increased radish biomass relative to the deionized-water control, with the highest total dry biomass recorded under industrial-effluent fertigation in the Siliceous Riparian Sandy Entisol. In contrast, separate analyses of leaf elemental concentrations detected significant Soil × Water interactions for cadmium, copper, magnesium, manganese, lead and zinc. Foliar sodium reached 25.7 g kg⁻¹ dry matter under landfill-leachate fertigation in the Calcareous Sandy Calcisol, indicating a potential crop-quality concern. Although the synthetic waters enhanced short-term biomass production, they also promoted substantial leaching of nutrients and salts. Therefore, their agronomic use in sandy soil requires careful control of water composition, application rate and drainage, together with validation using real wastewater in longer-term and field-scale experiments.

Keywords Wastewater · Fertigation · Sandy soils · Radish · Inorganic contaminants leaching · Groundwater quality · Salinity.



1 Introduction

The growing global demand for freshwater has been confronted with an increasingly limited availability of this resource. Projections indicate that by 2030, demand could far exceed the total available volume under sustained population growth, urban expansion, and increasing pressure on production systems (Ricart and Rico 2019). This water stress has been aggravated by the degradation of surface water as well as groundwater quality, which increases the vulnerability of agricultural systems that depend on good-quality water resources to maintain stable and safe yields (Sidhu et al. 2019). Competition among the agricultural, industrial, and domestic sectors for access to water resources has become a central challenge for sustainable water management and has driven the search for innovative strategies for saving, recycling, and reuse (Jeong et al. 2020). In this scenario, the reuse of treated wastewater has been considered a key strategy to relieve pressure on conventional sources and ensure crop irrigation. In particular, the use of treated domestic wastewater offers an additional and relatively stable source of irrigation water, less dependent on seasonal droughts and closely associated with urban flows (Gava et al. 2023).

In this context, the problem is exacerbated in sandy soil, considered particularly fragile for agricultural production due to its low water retention capacity, limited nutrient reserves, and reduced organic matter content (Alessandrino et al. 2021; de Holanda et al. 2025). The usage of treated wastewater as fertigation emerges as a particularly useful tool because it makes it possible to take advantage of dissolved nutrients present in effluents from wastewater treatment plants (WWTPs) and reduce dependence on external mineral fertilizers, if water quality remains within safe ranges for soil and crops (Muscarella et al. 2024). However, fertigation with treated wastewater poses sanitary, environmental, and regulatory challenges. These effluents may contain high concentrations of nutrients, metals, organic matter, and other compounds capable of triggering environmental issues (Mainardis et al. 2022; Chauhan and Kumar 2020). Therefore, their agricultural use requires both effective treatments and strict compliance with specific regulatory frameworks, including European directives on water quality and groundwater protection, complemented by the guidelines of the Food and Agriculture Organization (FAO) of the United Nations for wastewater reuse in agriculture (Radini et al. 2023; FAO 1992). From environmental, policy and economic perspectives, fertigation with treated effluents supports circular economy strategies by enhancing resource efficiency, promoting nutrient recycling and reducing dependence on mineral fertilisers, particularly under rising fertiliser costs. However, when application rates are not aligned with crop demand and soil retention capacity, drainage waters may transport mobile nutrients and salts beyond the root zone, increasing the risk of groundwater contamination (Azad et al. 2020). Therefore, the current literature advocates a holistic approach to assessing nutrient dynamics in soil, plant uptake, and corresponding drainage water quality (Albdaiwi et al. 2022; Romero-Trigueros et al. 2021). Even though the number of case studies is increasing, there is currently insufficient experimental evidence from controlled conditions to establish a clear connection among irrigation water quality, crop response, drainage composition, and soil nutrient dynamics. However, the abatement of major inorganic contaminants and trace elements (TEs) to meet FAO guidelines for irrigation water is often unfeasible. Therefore, alternative agronomic practices must be adopted to mitigate the impacts of using waters with concentrations exceeding the suggested limits (Obijanya et al. 2025).

Despite the increasing number of studies on wastewater fertigation, the marked variability in wastewater composition, treatment level and experimental conditions limits direct comparison among individual case studies. This study uses the evidence synthesized by Osorio et al. (2025) from 75 wastewater-fertigation studies to define standardized experimental water profiles and establish a consistent basis for evaluating the effects of different wastewater compositions. Category-specific median compositions were used to construct synthetic waters representing domestic wastewater, industrial effluent, and landfill leachate. This evidence-based approach reduced the influence of extreme concentrations and site-specific or temporal variability, allowing the effects of three contrasting water compositions to be compared under controlled conditions across the soil-plant-drainage system. Unlike studies based on a single wastewater source, this design provides a reproducible framework for examining how literature-derived wastewater profiles interact with nutrient-poor sandy soils.

Accordingly, this study aimed to evaluate the short-term response of the soil-plant-drainage system to fertigation with synthetic solutions reproducing treated domestic wastewater (DW), industrial effluent (IE), and landfill leachate (L), including compositions exceeding selected FAO irrigation-water quality guidelines (FAO 1992). Batch tests and a 40-day laboratory pot experiment with radish were conducted using two nutrient-poor sandy soils, a Siliceous Riparian Sandy Entisol (SRSE) and a Calcareous Sandy Calcisol (CSC). The study jointly evaluated nutrient and salt mobility, plant biomass, elemental accumulation in leaves and roots, post-experimental soil chemistry, and the temporal composition of drainage water under controlled laboratory conditions.

2 Materials and Methods

2.1 Soil and Fertigation Water Preparation

Two soils with different textures and chemical properties were selected to represent different pedological conditions (see Table 1): a Siliceous Riparian Sandy Entisol (SRSE) and a Calcareous Sandy Calcisol (CSC). Each soil was oven-dried at 80 °C for 24 h, gently disaggregated, and sieved through a 2-mm sieve to remove coarse fragments and plant debris. The homogenized material was stored in sealed polyethylene containers until use. The synthetic fertigation waters were formulated from the database compiled by Osorio et al. (2025), which included 75 wastewater-fertigation studies. Reported ion concentrations were standardized and grouped by water source, including domestic wastewater (28 studies), industrial effluents (21) and leachates (8). The source datasets were heterogeneous: the industrial-effluent category included different origins and treatment levels, whereas leachate compositions showed high dispersion and lacked consistent treatment reporting. Category-

Table 1 Basic physicochemical properties of the soils used in the tests

Soil	pH (H ₂ O)	EC _e (mS m ⁻¹)	Clay (%)	Sand (%)	Silt (%)	BD* (g cm ⁻³)	CEC (cmolc kg ⁻¹)	Org. C (g kg ⁻¹)	Tot. N (g kg ⁻¹)
CSC	7.7±0.1	0.26±0.02	5.0	90.0	5.0	1.38±0.02	2.6	5.8	<0.1
SRSE	7.7±0.1	0.20±0.04	10.0	75.5	14.5	1.47±0.02	2.5	0.6	<0.1

*BD: bulk density. Values are reported as mean±standard deviation

specific medians were therefore used instead of arithmetic means to reduce the influence of extreme observations. Indeed, when a dataset is skewed or contains extreme outliers, the median is preferable to the mean because it reliably represents the usual value without being influenced by extremely high or low numbers (Filzmoser et al. 2009). For example, the mean SO_4^{2-} and Na^+ concentrations reported for leachates (658.0 and 573.3 mg L^{-1} , respectively) were substantially higher than the corresponding medians (307.2 and 345.0 mg L^{-1}). The resulting profiles represent literature-derived central compositions, not typical European values or worst-case scenarios. Target concentrations were converted to molar units and reproduced using analytical-grade salts, as detailed in Table 1s. Deionized water was used as a rainwater proxy (B), and no real wastewater was used.

2.2 Pot Tests

Pot experiments were conducted in a controlled laboratory to evaluate the response of *Raphanus sativus* L. (radish) to fertigation with different water types. Polypropylene pots measuring 25 cm in height and 9.9 cm in internal diameter were used following Kang and Wan (2005), who recommended a minimum depth of 20 to 25 cm to provide sufficient space for root system development. The experiment followed a 2×4 factorial designs comprising two soil types, SRSE and CSC, and four fertigation water treatments, B, DW, IE and L. Two independent pots were established for each soil-water combination, resulting in 16 experimental units. Five seeds were initially sown in each pot to ensure successful plant establishment. After seedling emergence, the plants were thinned to one individual per pot. Therefore, each pot containing one plant was considered an independent experimental unit and biological replicate. Laboratory temperature was maintained at 20 °C throughout the experiment, while relative humidity was maintained at 60%. Fertigation was conducted for approximately 40 days under laboratory conditions. Irrigation requirements were estimated using the FAO CROPWAT model based on reference evapotranspiration and crop parameters, including the crop coefficient (Kc) and root depth (Allen et al. 1998). Irrigation was scheduled using a soil water balance approach to maintain soil moisture near field capacity, with irrigation triggered at 60% of critical depletion and assuming a 70% application efficiency. Fertigation was applied in 13 events between days 1 and 34, at intervals of 2 to 4 days depending on the crop developmental stage. The volume applied per event ranged from 68.26 to 205.60 mL per pot, resulting in a cumulative irrigation volume of 1744.84 mL per pot. The complete irrigation schedule is provided in Table 2s of the SI.

After each fertigation cycle, drainage water was collected, filtered through 0.45 μm membranes, and analyzed for pH and EC using a multiparameter probe (HI98194, Hanna Instruments, USA). Samples were stored in HDPE vials at 4 °C until analysis of DOC, major ions, and TEs. At the end of the tests, soil profiles in each pot were carefully separated into two depth intervals (0–10 and 10–20 cm) to characterize the vertical distribution of major ions and TEs (See section 3s in SI). The plants were harvested and separated into aboveground biomass, corresponding to the leaves, and belowground biomass. The entire belowground fraction was collected, weighed and processed as root biomass because the hypocotyl was not separated or quantified as an independent plant fraction. Fresh biomass was therefore determined separately for leaves and roots before further processing. Leaf material was cut with sterile scalpels, damaged tissue was removed, and the fresh biomass was recorded. Samples were placed in pre-weighed paper bags and then dried at 65 °C in a

ventilated oven for 24 h until a constant mass was reached, and reweighed. The dried leaves were ground using a laboratory mill, with rigorous cleaning between treatments to avoid cross-contamination.

The root systems were gently washed with running water to remove adhering soil particles, dried with absorbent paper, and weighed. They were then immersed in a 5 mM CaCl_2 solution for 15 min, rinsed with distilled water, and dried again. Root samples were placed in pre-weighed paper bags, dried at 65 °C until a constant mass was reached, and reweighed. The dried material was ground in the same way.

2.3 Analytical Methods

DOC in water was analysed measuring the ultraviolet absorbance at 254 nm (Weishaar et al. 2003) using an UV/VIS ONDA TOUCH UV-21 spectrophotometer. Major ions were analysed with an Inuvion Thermo Fisher Scientific chromatography system equipped with an isocratic dual pump with an IonPac AS14A 4 × 250 mm column, the pre-column AG14A 4 × 50 mm, and an ADRS 600 (4 mm) self-suppressor for anions and an IonPac CS12A 4 × 250 mm column, the pre-column CS12A 4 × 50 mm thermostatted at 35 °C, and a CDRS 600 (4 mm) self-suppressor for cations. The AS-DV Thermo Fisher Scientific autosampler was employed to run the analysis, while quality control (QC) samples were run every 20 samples. TEs were analysed with an ICP-OES (PerkinElmer, USA) to quantify their concentration in water samples after acidification with ultrapure 1 M nitric acid and filtering on 0.45 µm. Element concentrations were additionally determined by ICP-OES using a second subsample analysed with a hydride generation system to minimize spectral interferences. All the equipment was correctly acid washed, and method blanks were carried out. For all plant samples, dry matter content was determined gravimetrically prior to microwave-assisted digestion ($\text{HNO}_3\text{:H}_2\text{O}_2$, 10:1) and subsequent determination of total element concentrations by ICP-OES (iCAP PRO XP Duo, Thermo Fisher Scientific). Analytical quality control included reagent blanks, duplicate samples and certified reference plant materials from the interlaboratory calibration programme (WEPAL-ISE).

2.4 Statistical Analysis

Statistical analyses were selected according to the structure of each dataset. For endpoint variables measured once per experimental unit, analysis of variance was used. Specifically, two-way ANOVA was performed to evaluate the main and interactive effects of soil type, SRSE and CSC, and fertigation water treatment, B, DW, IE and L, on plant biomass, leaf elemental composition and post-experimental soil chemistry. The pot was considered the experimental unit, with two independent pots per Soil × Water combination. Biomass data were log10-transformed before analysis to reduce right-skewness and stabilize variance across treatments. When significant differences were detected, Tukey's HSD test was used for pairwise comparisons. Because separate ANOVAs were conducted for multiple response variables, p-values were adjusted using the Benjamini-Hochberg procedure to control the false discovery rate, separately for the Soil, Water and Soil × Water terms.

Root elemental composition was evaluated descriptively because only one analytical result was available for each Soil × Water combination, preventing a valid inferential analysis. Drainage-water variables were analyzed using linear mixed-effects models, with Soil,

Water, Time and their interactions included as fixed effects and pot identity included as a random effect to account for repeated measurements. The significance of the fixed effects, including the Soil $\times$ Water $\times$ Time interaction, was evaluated using Wald chi-square tests. P-values for the drainage-water models were adjusted across the 14 response variables using the Benjamini-Hochberg procedure. Because only two independent pots were available per Soil $\times$ Water combination, statistical inference was considered exploratory and restricted to the experimental conditions. This limited replication reduced the strength of the approach, particularly for detecting interactions. Repeated drainage-water measurements improved temporal resolution but did not increase biological replication; therefore, p-values were interpreted alongside effect direction, magnitude and consistency, using $p < 0.05$ as the threshold for statistical significance. All analyses were performed using Python 3.14.6, pandas 3.0.5, SciPy 1.18.0 and statsmodels 0.14.6.

3 Results and Discussion

3.1 Water Leaching Dynamics

Drainage-water composition showed marked temporal variation among the experimental treatments. Linear mixed-effects models revealed significant Soil $\times$ Water $\times$ Time interactions for 13 of the 14 evaluated responses after Benjamini-Hochberg adjustment (Table 2). PO_4^{3-} was the only variable for which the three-way interaction was not significant. The temporal trajectories of most drainage-water parameters varied according to the specific Soil $\times$ Water combination, supporting their interpretation at the treatment-combination level rather than through the separate main effects of Soil and Water. Given the limited biological replication, these interactions should be interpreted as evidence of different temporal patterns within the present experiment rather than as precise estimates

Table 2 Mixed-effects model results for the Soil $\times$ Water $\times$ Time interaction in drainage-water composition during the pot experiment

Response	Wald χ^2	Raw p -value	Adjusted p -value	Significance
EC	45.591	<0.001	<0.001	***
pH	41.094	<0.001	<0.001	***
DOC	27.009	0.008	0.009	**
NH_4^+	152.243	<0.001	<0.001	***
Mg^{2+}	28.762	0.004	0.005	**
NO_3^-	128.350	<0.001	<0.001	***
Cl^-	60.842	<0.001	<0.001	***
Cu	66.013	<0.001	<0.001	***
Zn	54.302	<0.001	<0.001	***
Fe	24.282	0.019	0.020	*
Cr	388.025	<0.001	<0.001	***
Na^+	80.516	<0.001	<0.001	***
PO_4^{3-}	18.248	0.108	0.108	NS
K^+	189.223	<0.001	<0.001	***

Wald chi-square tests evaluate whether the temporal response differed among Soil $\times$ Water combinations. P-values were adjusted across the 14 drainage-water responses using the Benjamini-Hochberg procedure. *, ** and *** indicate significance at $p < 0.05$, $p < 0.01$ and $p < 0.001$, respectively; NS, not significant

of treatment-specific trajectories. The repeated observations improved characterization of temporal variation but did not compensate for the small number of independent pots.

Both SRSE and CSC behaved as low-retention, low-CEC matrices with minimal organic carbon (Figs. 1 and 2), consistently with the batch-test results (see Section 4s in the SI), resulting in limited nutrient retention. The significant interactions observed for most

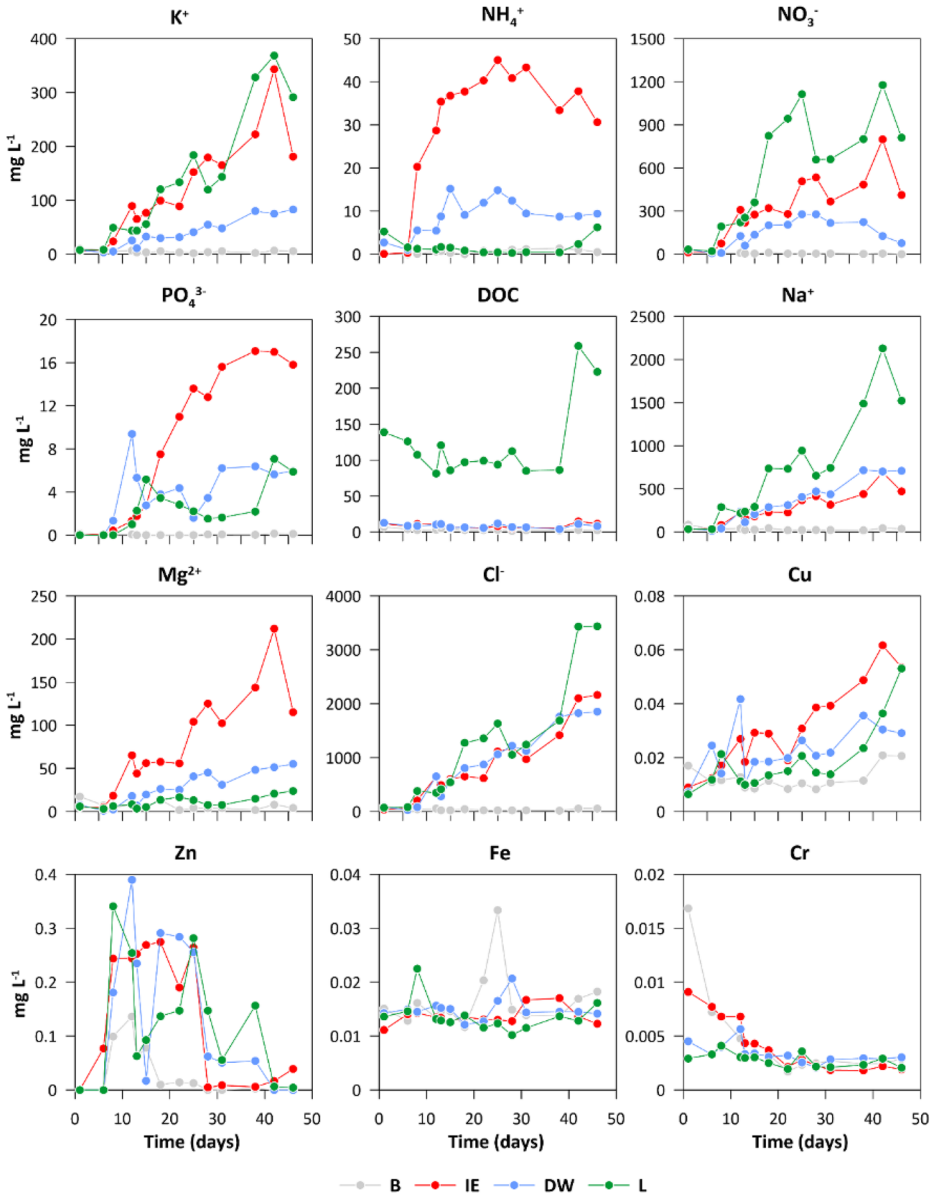


Fig. 1 Temporal variation of dissolved major ions and trace elements (TEs) in drainage water during the pot test for CSC. Abbreviations: B, blank control with deionized water; IE, industrial effluent; DW, domestic wastewater; L, landfill leachate; CSC, Calcareous Sandy Calcisol; TEs, trace elements; DOC, dissolved organic carbon

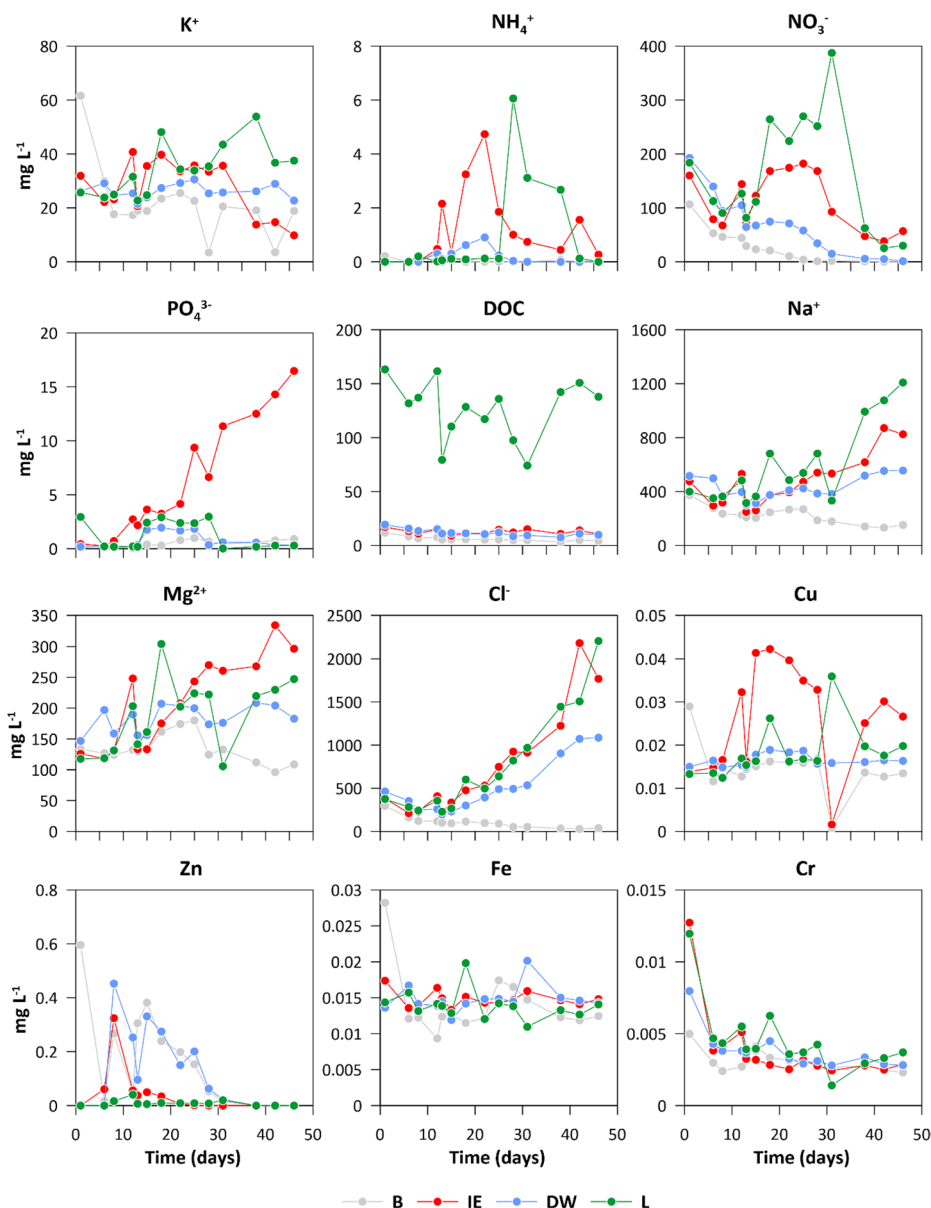


Fig. 2 Temporal variation of dissolved major ions and trace elements (TEs) in drainage water during the pot test for SRSE. Abbreviations: B, blank control with deionized water; IE, industrial effluent; DW, domestic wastewater; L, landfill leachate; SRSE, Siliceous Riparian Sandy Entisol; TEs, trace elements; DOC, dissolved organic carbon

drainage-water variables further showed that the magnitude and timing of solute transport differed among the experimental combinations. Consequently, the collected leachates frequently exceeded the EU Groundwater Directive, 2006/118/EC thresholds. Maximum mean concentrations of the principal solutes in drainage water are summarized in Table 3 of the

SI. The highest NO_3^- , Cl^- , Na^+ and K^+ concentrations were recorded under L-CSC on day 42, reaching 1177.87, 3428.66, 2129.86 and 368.56 mg L^{-1} , respectively. The maximum NH_4^+ concentration was 45.03 mg L^{-1} under IE-CSC on day 25, whereas PO_4^{3-} reached 17.09 mg L^{-1} under IE-CSC on day 38.

NO_3^- concentrations closely followed the composition of the fertigation waters and the observed drainage patterns, suggesting that input concentration and water-driven transport were important under the experimental conditions. Across both soils, NO_3^- exhibited a near-conservative behaviour over most of the test period. In CSC pots under IE and L, NO_3^- rapidly exceeded the 50 mg L^{-1} threshold. However, NO_3^- concentrations in the leachates declined markedly toward the end of the test, indicating that most of the readily mobile nitrate had likely been leached earlier in the experiment. This decline could also be associated with increasing plant N demand during the final growth phase (Solangi et al. 2024), although plant N uptake was not directly quantified. Furthermore, the lower concentrations observed in SRSE may reflect soil-specific differences in pore-water residence time and microbial N transformations (Gu and Riley 2010). However, pore-water residence time, denitrification rates and microbial activity were not measured; therefore, the contribution of these processes cannot be determined. Nevertheless, these patterns are consistent with the high mobility of NO_3^- generally reported in coarse-textured soils with low sorption capacity, but do not demonstrate a specific transport or microbial mechanism (Chen et al. 2024). Overall, NO_3^- leaching emerged as the dominant environmental risk, with clear implications for groundwater quality and eutrophication potential (Sanchezperez et al. 2003).

Rubio-Asensio and Intrigliolo (2024) similarly found lower NO_3^- concentrations in the root zone when fertigation frequency was increased, despite identical weekly water and fertilizer inputs, highlighting the importance of synchronizing water application with crop N demand to limit nitrate leaching. Conversely, NH_4^+ exhibited a clear soil-dependent behaviour. In CSC, fertigation produced early NH_4^+ peaks above the 0.5 mg L^{-1} threshold. In contrast, NH_4^+ peaks in SRSE were modest and displayed a periodic pattern. The higher NH_4^+ concentrations in CSC leachates may reflect reduced retention under high ionic-strength conditions, as indicated by the EC trends in Figs. 2s and 3s of the SI. Competition with Ca^{2+} and Mg^{2+} for exchange sites can increase NH_4^+ mobility (Jellali et al. 2010), while the low organic carbon content of SRSE and CSC may further limit sorptive and microbially mediated retention. This interpretation is consistent with Garbowski (2024), who observed greater nutrient retention and lower N losses after incorporating biologically active algal biomass into sandy soil. Nevertheless, microbial processes were not directly quantified in the present study. These processes should therefore be regarded as possible explanations rather than demonstrated causes of the observed soil-specific patterns.

The PO_4^{3-} behaviour was consistent with the batch tests (see Section 4s and Fig. 1s in the SI), showing a marked decrease relative to the fertigation inputs, which is consistent with retention by reactive soil surfaces. However, under irrigation with IE, apparent increases in PO_4^{3-} concentrations were observed during the later sampling dates. This trend could reflect a progressive reduction in retention efficiency under high P loading, potentially involving saturation of high-affinity sorption sites rather than complete removal from porewater (Van Der Salm et al. 2016). However, sorption-site occupancy and mineral precipitation were not measured, so the mechanism responsible for this pattern cannot be confirmed. Moreover, the non-significant Soil $\times$ Water $\times$ Time interaction indicates that

this temporal response was not sufficiently consistent across the experimental units. Na^+ concentrations in the leachates were high and closely reflected the Na^+ inputs, particularly in CSC under L fertigation. This conservative behavior highlights a potential long-term concern, as prolonged fertigation with Na-rich waters may increase Na^+ loading to groundwater and, if sustained over time, could pose a risk to soil structural stability (Gao et al. 2021). Cl^- showed conservative behavior throughout the test, with leachates closely reflecting fertigation inputs (Zhao et al. 2025). However, Cl^- concentrations increased gradually, which could be associated with concentration effects arising from plant water uptake and the observed decline in soil moisture during the later developmental stages (Alessandrino et al. 2024; Figs. 2s and 3s in the SI). Actual evapotranspiration and pore-water residence time were not directly measured, so the contribution of evapoconcentration cannot be quantified. Hydraulic loading should be adjusted to both soil moisture conditions and the hydraulic capacity of the receiving soil. He et al. (2020) showed that regulating wastewater application according to soil-water conditions reduced the risk of hydraulic overloading, emphasizing the need to align fertigation rates with crop demand and soil drainage capacity. However, porosity, saturated hydraulic conductivity and soil drainage capacity were not measured in the present experiment. Consequently, the observed drainage patterns cannot be used to demonstrate hydraulic overloading or differences in hydraulic functioning between the two soils. In addition, the potential effects of Na^+ cannot be assessed from salinity alone, as soil structural responses depend on the relative abundance of monovalent and divalent cations. Xie et al. (2024) found that decreasing the sodium adsorption ratio and increasing Ca^{2+} availability mitigated aggregate deterioration under brackish-water irrigation. A similar mechanism may underlie the structural risks associated with sustained Na^+ loading in the studied soils. Nevertheless, this mechanism could not be verified because aggregate stability, porosity and hydraulic conductivity were not measured. Therefore, no treatment-induced change in soil structure or hydraulic functioning can be inferred from the present results.

The temporal variation of K^+ concentrations in drainage water, including the late-stage increase observed in CSC, could have resulted from a combination of solution-phase transport and short-term ionic interactions. Classical cation exchange may have been limited by the low CEC and coarse texture of both soils. The parallel trends of K^+ and Mg^{2+} may thus reflect transient competitive interactions within the soil solution rather than extensive exchange on mineral surfaces (Baki et al. 2025). In CSC, the late-stage increase in K^+ concentrations could reflect changes in ionic equilibria or transport conditions rather than simple unidirectional sorption. Moreover, irrigation with high-ionic-strength waters rich in Na^+ may enhance cation mobilization, as Na^+ can compete for exchange sites and promote the displacement of previously retained cations (Hasan and Abu-Awwad 2025). However, exchange-site occupancy and cation-exchange dynamics were not measured, so these processes remain possible explanations. The temporal profiles of DOC differed among fertigation treatments, with the highest mean concentration recorded under L-CSC on day 42 at 12.94 mg L^{-1} , but no consistent difference was observed between the two soils. However, DOC was consistently lower than the corresponding inputs, which is consistent with DOC retention or transformation within the soil system, as already observed in the batch tests. Physicochemical processes such as cation bridging and adsorption onto reactive mineral surfaces could have contributed to DOC removal (Xu and Tsang 2024). However, microbial activity and DOC biodegradation were not quantified; therefore, the relative contributions

of physicochemical retention and biological transformation cannot be determined. Moreover, the similarity between DOC and PO_4^{3-} trends suggests coupled removal processes, potentially involving co-precipitation or joint retention mechanisms (Gerke 2010). This interpretation remains hypothetical because these processes were not measured directly. In longer-term systems, biological accumulation may modify the hydraulic and treatment performance of sandy media. Wu et al. (2021) showed that biomass accumulation in intermittent sand filters contributed to contaminant removal while simultaneously modifying permeability through bioclogging. Although microbial biomass and hydraulic changes were not directly measured in the present short-term experiment, bioclogging should be considered a hypothesis for future longer-term investigation rather than an explanation of the drainage patterns observed here.

TE concentrations in leachates remained consistently low throughout the test (Figs. 1 and 2). All measured concentrations of Fe, Cr, Cu and Zn were far below thresholds relevant to groundwater protection, remaining well under WHO health-based guideline values for total Cr (0.05 mg L^{-1}) and Cu (2.0 mg L^{-1}), and below aesthetic limits for Fe (0.3 mg L^{-1}) and Zn (3 mg L^{-1}) (WHO 2017). Even for Cu and Zn, which were directly supplied with the fertigation waters, leachate concentrations remained well within safe limits, while Fe and Cr variations were minor and linked to soil-derived fractions or isolated redox fluctuations. Although the Soil $\times$ Water $\times$ Time interactions were significant for all four TEs, this statistical significance reflected differences in their temporal trajectories rather than concentrations exceeding environmental thresholds. Recent field evidence also indicates that TE transfer under wastewater irrigation is strongly influenced by soil properties and plant-specific accumulation, rather than by wastewater concentrations alone (Ibrahim and Hammad 2024). The low concentrations measured in drainage water, therefore, indicate limited short-term leaching of the evaluated TEs under the tested conditions. This finding is consistent with previous studies reporting strong immobilization of Cu and Zn in wastewater-irrigated soils, where pH, carbonates and organic matter promote their retention in low-mobility forms (Kovač et al. 2022; Abd-Elwahed 2018). However, the present data do not allow the relative contributions of soil retention and plant uptake to be quantified. Under the low metal loads tested, TEs did not represent a significant short-term environmental concern. Instead, the main risks to groundwater quality were associated with salinity and nutrient loads, consistent with observations from similar systems (Qadir et al. 2006). Post-experimental soil chemistry was primarily influenced by fertigation water composition. After Benjamini-Hochberg adjustment, a significant water effect was detected for K^+ [$F(3,8)=8.34$, adjusted $p=0.033$], Na^+ [$F(3,8)=18.78$, adjusted $p=0.018$] and Cl^- [$F(3,8)=8.93$, adjusted $p=0.033$]. No significant effects were detected for NH_4^+ , NO_3^- , PO_4^{3-} , Fe, Cr or Cu, and the Soil $\times$ Water interaction was not significant for any of the evaluated variables. Zn was excluded from inferential analysis because all post-experimental soil concentrations were below the analytical quantification limit and were therefore recorded as zero, resulting in no variability among samples. Complete ANOVA results are provided in Table 4s of the SI. The drainage-water and post-experimental soil results indicate different pathways for the applied solutes. Highly mobile constituents were mainly reflected in their temporal losses through drainage, while K^+ , Na^+ and Cl^- also showed a detectable treatment effect in the soil at the end of the experiment. Therefore, the absence of a significant final soil effect for the other analytes does not necessarily indicate limited transport, as part of the applied load may have already been lost through drainage.

3.2 Radish Response to Fertigation

3.2.1 Biomass Response

Radish biomass differed markedly among fertigation waters and between the two soils (Table 3). The nutrient-containing waters promoted greater leaf and root development than the blank treatment, although the magnitude of the response varied among biomass components.

Two-way ANOVA showed significant effects of Soil [$F(1,8)=6.05$ – 73.74 , adjusted $p \leq 0.039$] and Water [$F(3,8)=11.66$ – 54.63 , adjusted $p \leq 0.003$] on all biomass variables. In contrast, the Soil $\times$ Water interaction was not statistically significant [$F(3,8)=0.47$ – 2.20 , adjusted $p=0.453$ – 0.708]. Interaction effects were specific to each response variable and statistical model. Therefore, the non-significant Soil $\times$ Water interaction for biomass does not conflict with the significant Soil $\times$ Water interactions detected for some leaf elemental concentrations or with the significant Soil $\times$ Water $\times$ Time interactions observed for several drainage variables. These results indicate that treatment-dependent responses were detected for certain chemical variables, but not for biomass under the present experimental design. Given the limited biological replication, this result indicates only that no Soil $\times$ Water interaction was detected under the present experimental design and should not be interpreted as evidence that biomass responses to fertigation water were equivalent between the two soils.

The significant soil effect was reflected in the generally greater biomass obtained in SRSE, particularly for healthy leaf fresh mass, clean root fresh mass and total dry biomass. Across fertigation waters, IE produced the greatest biomass, with the highest total dry biomass recorded in IE-SRSE at 5.55 ± 0.21 g. By comparison, total dry biomass remained below 0.5 g in both blank treatments. DW and L also increased biomass relative to B, although the magnitude of this increase differed among plant fractions. Although numerical differences were observed among individual Soil $\times$ Water combinations, no interaction was detected under the present design. Given the limited biological replication, this result should not be interpreted as evidence of equivalent biomass responses between the two soils. The significant Soil main effect indicates that biomass production differed between SRSE and CSC, but the measurements performed in this experiment do not identify the

Table 3 Biomass production of radish under the different Soil $\times$ Water combinations

Treatment	Healthy leaf fresh mass (g)	Clean root fresh mass (g)	Leaf dry mass (g)	Root dry mass (g)	Total dry biomass (g)
B-SRSE	$2.13 \pm 1.52b$	1.48 ± 0.57 cd	$0.20 \pm 0.14b$	$0.25 \pm 0.07bc$	$0.45 \pm 0.21c$
IE-SRSE	$45.14 \pm 6.87a$	$10.38 \pm 0.11a$	$3.95 \pm 0.07a$	$1.60 \pm 0.14a$	$5.55 \pm 0.21a$
DW-SRSE	$15.00 \pm 8.49a$	$6.75 \pm 2.08a$	$1.45 \pm 1.20a$	$0.95 \pm 0.07ab$	$2.40 \pm 1.13ab$
L-SRSE	$30.50 \pm 3.54a$	$4.52 \pm 1.10ab$	$2.55 \pm 0.49a$	$0.60 \pm 0.28abc$	$3.15 \pm 0.21ab$
B-CSC	$0.64 \pm 0.01b$	$0.55 \pm 0.21d$	$0.15 \pm 0.07b$	$0.20 \pm 0.14c$	$0.35 \pm 0.07c$
IE-CSC	$15.00 \pm 0.00a$	$2.22 \pm 0.69bc$	$1.45 \pm 0.21a$	$0.35 \pm 0.07bc$	$1.80 \pm 0.28b$
DW-CSC	$11.50 \pm 4.95a$	$2.20 \pm 0.42bc$	$0.70 \pm 0.14ab$	$0.55 \pm 0.07abc$	$1.25 \pm 0.07b$
L-CSC	$11.00 \pm 0.00a$	$1.50 \pm 0.21bcd$	$1.35 \pm 0.07a$	$0.30 \pm 0.00bc$	$1.65 \pm 0.07b$

Values are mean $\pm$ SD of two independent pots. Different letters indicate significant differences among Soil $\times$ Water combinations (Tukey HSD, $p < 0.05$). Analyses used log10-transformed data, while untransformed means are shown. Total dry biomass was calculated as leaf plus root dry mass. *B* blank control with deionized water, *IE* industrial effluent, *DW* domestic wastewater, *L* landfill leachate, *SRSE* Siliceous Riparian Sandy Entisol, *CSC* Calcareous Sandy Calcisol

mechanism responsible for this difference. However, Li et al. (2024) found that stable soil-water conditions increased cherry radish biomass and enhanced N, P and K acquisition compared with fluctuating water conditions. Their findings suggest that differences in soil-water availability could contribute to soil-dependent growth responses. Nevertheless, water retention, pore-water residence time and hydraulic behaviour were not directly quantified in the present experiment. These processes therefore remain possible explanations and cannot be considered demonstrated causes of the greater biomass observed in SRSE. The contrasting growth patterns among treatments were also visually apparent in the representative plants photographed at harvest (Fig. 3).

These biomass responses confirmed the strong influence of nutrient supply. Fertigation with waters presenting higher ionic and nutrient loads, particularly IE and L, markedly increased aboveground and belowground biomass relative to the control, consistent with the agronomic benefits of wastewater-derived nutrients reported by Libutti et al. (2017). However, the increase in biomass occurred simultaneously with substantial nutrient and salt transport in drainage water. Thus, the short-term agronomic response did not necessarily correspond to a reduction in the potential environmental risk associated with fertigation. Onchoke and Fateru (2023) similarly attributed increased radish biomass in substrates amended with wastewater biosolids to their N and P supply. However, they also found that elemental distribution among plant organs varied with both the element and substrate composition. Therefore, the biomass gains observed here should be evaluated alongside solute losses through drainage and elemental accumulation in plant tissues. Although total biomass increased, development of the harvestable storage organ remained limited. A clearly differentiated swollen hypocotyl was not observed in most pots. Because the hypocotyl was not separated or quantified independently at harvest, the entire belowground fraction was reported as root biomass and no separate numerical dataset for hypocotyl biomass was generated. Hypocotyl thickening is particularly sensitive to the osmotic status and ionic balance of the rhizosphere because it depends on the maintenance of an adequate water gradient and the efficient partitioning of assimilates towards storage tissues (Noman et al. 2017). In this context, salinity stress and elevated soil electrical conductivity may alter biomass allocation and radish growth, thereby affecting the development and quality of the harvestable organ

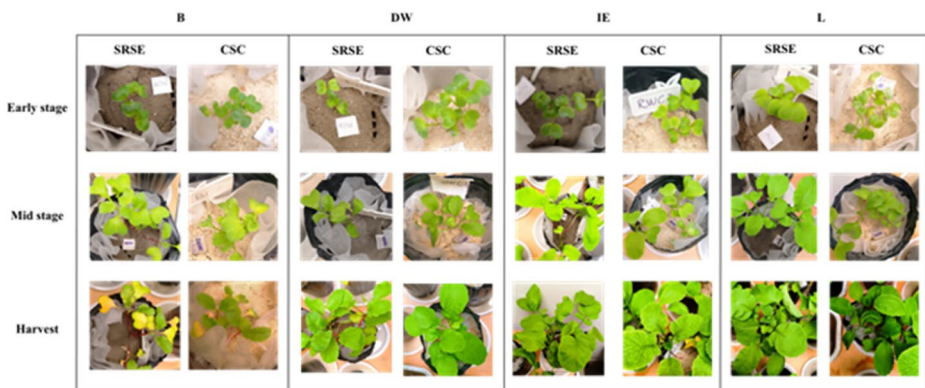


Fig. 3 Representative images of radish plants grown under the different fertigation water and soil types at the end of the test. Abbreviations: B, blank control with deionized water; IE, industrial effluent; DW, domestic wastewater; L, landfill leachate; SRSE, Siliceous Riparian Sandy Entisol; CSC, Calcareous Sandy Calcisol

even in the absence of visible foliar toxicity during early growth stages (Zhu et al. 2024). Therefore, the greater total biomass observed under IE, DW and L should not be interpreted as evidence of successful development of the marketable storage organ.

3.2.2 Elemental Composition of Leaves and Roots

Two-way ANOVA identified significant Soil $\times$ Water interactions for Cd, Cu, Mg, Mn, Pb and Zn [$F(3, 8)=5.74$ to 19.38 , adjusted $p=0.007$ to 0.047], showing that the response of these elements to fertigation water differed between SRSE and CSC. For Ca, K, Na, Ni, P and S, the interaction was not significant, whereas both Soil [$F(1, 8)=7.48$ to 55.91 , adjusted $p\leq 0.033$] and Water [$F(3, 8)=12.17$ to 42.49 , adjusted $p\leq 0.003$] exerted significant main effects. Fe was not significantly affected by Soil, Water or their interaction. Complete ANOVA results are provided in Table 5s of the SI. Leaf macronutrient concentrations (Table 4) generally fell within the sufficiency ranges reported for radish (Gamba et al. 2021). Among these elements, Mg showed the clearest soil-dependent response. Its highest concentration was recorded under IE in CSC, whereas the increase associated with the same fertigation water was less pronounced in SRSE. This interaction suggests that Mg uptake was influenced by the combined effects of Mg supply, soil properties and competition with other ions in the root-zone solution.

Foliar P remained in the lower sufficiency band, especially in blank treatments and occasionally in CSC, indicating a moderate P status consistent with restricted P availability despite the retention of phosphate within the soil system. However, the low foliar P high-

Table 4 Macronutrients, major ions, micronutrients and trace metals in radish leaves

Test	Macronutrients and major ions (mg kg ⁻¹)						
	Ca	K	Mg	Na	P	S	
<i>B- SRSE</i>	47,638±464	28,606±589	6300±98	3500±101	2215±115	14,310±112	
<i>IE- SRSE</i>	29,401±549	47,989±908	7300±45	11,300±234	4550±334	9599±102	
<i>DW- SRSE</i>	37,594±453	38,898±889	4700±56	11,600±432	2420±267	9171±103	
<i>L-SRSE</i>	34,359±329	39,522±342	5000±143	14,500±221	2664±99	9019±189	
<i>B-CSC</i>	53,753±678	22,271±267	4500±122	4900±223	1311±78	21,452±423	
<i>IE-CSC</i>	35,176±332	34,299±334	10,000±221	20,800±789	3576±69	12,220±208	
<i>DW-CSC</i>	38,506±534	37,554±445	5200±102	9300±104	1827±111	9984±76	
<i>L-CSC</i>	37,231±221	37,189±190	3700±80	25,700±453	3102±71	10,579±531	
Test	Micronutrients and trace metals (mg kg ⁻¹)						
	Cd	Cu	Fe	Mn	Ni	Pb	Zn
<i>B- SRSE</i>	0.2±0.1	3.3±0.2	211.0±9.3	178.0±6.8	2.1±0.2	0.8±0.1	33.1±1.1
<i>IE- SRSE</i>	<0.070	5.6±0.3	129.0±4.5	133.0±7.9	1.6±0.1	<0.50	38.8±0.9
<i>DW- SRSE</i>	<0.070	2.8±0.2	88.2±1.4	140.0±8.8	1.2±0.3	<0.50	22.3±2.1
<i>L-SRSE</i>	0.2±0.0	3.7±0.3	104.0±9.7	123.0±9.0	0.9±0.1	<0.50	30.0±0.9
<i>B-CSC</i>	<0.070	93.0±1.9	103.0±6.5	119.0±8.9	3.7±0.4	1.2±0.2	38.2±0.7
<i>IE-CSC</i>	<0.070	44.1±0.9	52.5±2.1	130.0±6.4	3.7±0.5	<0.50	87.0±2.3
<i>DW-CSC</i>	0.1±0.0	21.1±0.3	57.3±3.4	155.0±6.6	1.9±0.2	<0.50	37.6±1.1
<i>L-CSC</i>	0.1±0.0	23.4±0.6	64.7±5.6	148.0±7.8	2.3±0.1	<0.50	32.8±0.7

Values are presented as mean $\pm$ standard deviation of two independent pots. Values preceded by "<" were below the analytical quantification limit. Significant effects of Soil, Water and/or their interaction were detected for all elements except Fe. *B* blank control with deionized water, *IE* industrial effluent, *DW* domestic wastewater, *L* landfill leachate; SRSE, Siliceous Riparian Sandy Entisol, CSC Calcareous Sandy Calcisol

lights that sole fertigation does not ensure efficient P cycling in sandy soils (Jalali and Jalali 2016; Liu et al. 2012). Tissue concentration and total plant acquisition should nevertheless be distinguished. Li et al. (2024) found that increased nutrient uptake in cherry radish resulted primarily from greater biomass rather than from higher tissue concentrations. Accordingly, the foliar values reported here describe treatment-induced changes in elemental composition but should not be interpreted as direct estimates of total nutrient uptake, which would require integrating concentration with the biomass of each plant fraction.

Foliar Na accumulation increased markedly under L and, to a lesser extent, under IE and DW, reaching 25.7 g kg^{-1} dry matter under L-CSC, equivalent to approximately 2.6% of dry matter (Table 4). This pattern was consistent with the elevated Na concentrations measured in drainage water (Figs. 1 and 2), reflecting substantial transfer of Na from the fertigation solutions to aboveground tissues. Accordingly, crop assessment under wastewater fertigation should not be restricted to potentially toxic elements, since substantial Na accumulation may also alter the chemical and organoleptic quality of edible tissues. Saline irrigation has been shown to modify soluble solids, sugars, proteins and vitamin C concentrations in radish (Zhu et al. 2024), while NaCl accumulation in edible leaves has been associated with a perceptible salty taste in sensory evaluations (Amoruso et al. 2022). However, sensory attributes were not evaluated in the present study, and the marketable hypocotyl was not analysed separately. Therefore, the observed foliar Na enrichment identifies a potential organoleptic and crop-quality concern rather than direct evidence of altered taste, consumer acceptability or commercial quality. Moreover, although plants retained part of the applied Na, repeated application of Na-rich waters may promote salinity- and sodicity-related deterioration of soil structure and hydraulic functioning, particularly in soils with limited buffering capacity (Kramer et al. 2022; Adeyemo et al. 2022). This represents a potential longer-term risk identified from previous studies and was not directly evaluated in the present experiment because aggregate stability, porosity and saturated hydraulic conductivity were not measured. Future studies should therefore combine tissue-specific Na measurements, sensory evaluations and direct monitoring of soil physical and hydraulic properties.

Most micronutrients remained within the ranges generally reported for vegetable tissues, although Zn enrichment was observed under IE-CSC without evidence of hyperaccumulation (Kabata-Pendias 2010). The Soil $\times$ Water interactions identified for several trace elements reflected differences in accumulation patterns rather than uniformly high concentrations. Cd and Pb were below or close to their analytical quantification limits in most treatments, while Ni remained at relatively low concentrations despite significant Soil and Water main effects. Therefore, the statistical responses of these elements did not correspond to substantial accumulation in leaf tissues. The pronounced interaction observed for Cu was driven principally by the high concentration measured in B-CSC and the lower concentrations recorded under treatments that produced greater biomass. Because this pattern did not correspond directly to the Cu supplied by the fertigation waters, soil-derived Cu and biomass dilution probably contributed more strongly than irrigation inputs alone. Zn also showed a soil-dependent response, with marked enrichment under IE-CSC but no comparable increase in SRSE. In contrast, Fe varied numerically among treatments but was not significantly affected by either experimental factor. Overall, the low Cd and Pb concentrations, together with the moderate concentrations of Fe, Mn, Ni and Zn, do not suggest substantial short-term accumulation of potentially toxic elements in leaf biomass. However, this interpretation is limited to the 40-day

experimental period and does not exclude cumulative effects from repeated wastewater application over longer cultivation periods (Libutti et al. 2017). Root composition was evaluated descriptively because only one analytical result was available for each Soil $\times$ Water combination, preventing factorial ANOVA and Tukey comparisons. Therefore, the values in Table 5 describe only numerical differences among treatments. Roots fertigated with IE and L generally showed higher K and P concentrations than those receiving DW, together with treatment-specific increases in Mg and S, consistent with the higher nutrient loads of these waters. However, these patterns cannot be considered evidence of significantly greater nutrient uptake and may reflect the combined effects of water composition, soil-controlled availability and element partitioning among plant organs. Ibrahim and Hammad (2024) similarly reported that elemental concentrations under treated wastewater irrigation varied with soil properties and crop-specific accumulation. Nevertheless, the lack of replicated root analyses prevented these mechanisms from being statistically distinguished in the present study. Ca concentrations were lower in SRSE than in CSC, although the highest SRSE value occurred under L. This numerical pattern may reflect differences in Ca availability and transpirational transport, but the proposed mechanism could not be evaluated inferentially with the available root dataset (White 2003). No root elemental composition data are presented for the B-SRSE treatment because root development was limited and the recovered biomass was insufficient for chemical analysis. Root P remained relatively low across treatments, whereas Na concentrations were numerically highest under IE and L, particularly in CSC. In *Raphanus*

Table 5 Macronutrients, major ions, micronutrients and trace metals in radish roots

Tests	Macronutrients and major ions (mg kg ⁻¹)						
	Ca	K	Mg	Na	P	S	
<i>B- SRSE</i>	NA	NA	NA	NA	NA	NA	
<i>IE- SRSE</i>	6.8±0.7	35,700±2200	2.4±0.4	7.0±0.8	4.4±0.2	6.8±0.3	
<i>DW- SRSE</i>	5.8±0.3	32,100±2800	1.6±0.2	6.2±0.9	2.3±0.1	6.0±0.4	
<i>L-SRSE</i>	11.1±0.9	37,400±3900	2.1±0.2	7.5±0.6	3.7±0.3	7.0±0.1	
<i>B-CSC</i>	69.6±5.4	27,700±1300	1.7±0.2	8.0±0.4	1.4±0.1	9.1±0.3	
<i>IE-CSC</i>	15.9±1.1	35,800±2200	2.5±0.4	13.8±0.1	2.7±0.2	7.9±0.4	
<i>DW-CSC</i>	30.0±2.1	30,700±2400	1.9±0.3	5.3±0.4	1.8±0.1	8.0±0.4	
<i>L-CSC</i>	53.5±3.4	30,900±2800	1.7±0.1	16.5±0.1	2.9±0.3	9.0±0.3	
Tests	Micronutrients and trace metals (mg kg ⁻¹)						
	Cd	Cu	Fe	Mn	Ni	Pb	Zn
<i>B- SRSE*</i>	NA	NA	NA	NA	NA	NA	NA
<i>IE- SRSE</i>	<0.20	12.4±0.8	1350±40.1	57.2±2.3	7.28±0.9	<0.70	32.9±2.1
<i>DW- SRSE</i>	<0.20	3.7±0.2	682±33.2	35.1±3.3	4.22±0.3	<0.70	18.2±2.2
<i>L-SRSE</i>	<0.20	17.2±0.1	359±23.9	33.9±2.1	3.00±0.4	<0.70	27.9±2.4
<i>B-CSC</i>	<0.2	54.0±2.3	744±56.4	104.0±6.5	5.55±0.8	<0.70	27.0±1.2
<i>IE-CSC</i>	<0.20	45.5±1.2	664±44.5	37.1±4.3	4.66±0.2	<0.70	49.6±3.5
<i>DW-CSC</i>	<0.20	22.5±1.1	276±11.2	42.9±3.3	2.84±0.1	<0.70	20.9±2.2
<i>L-CSC</i>	<0.20	22.1±0.9	459±11.3	58.5±3.6	2.57±0.3	<0.70	23.2±1.9

Root elemental concentrations are reported as provided by the analytical laboratory. Values preceded by "<" were below the analytical quantification limit. NA indicates that elemental composition was not determined because insufficient root material was available for chemical analysis. *B* blank control with deionized water, *IE* industrial effluent, *DW* domestic wastewater, *L*, landfill leachate, *SRSE* Siliceous Riparian Sandy Entisol, *CSC* Calcareous Sandy Calcisol

sativus, roots may act as transient Na sinks through cellular compartmentalization mechanisms (Libutti et al. 2017). Nevertheless, the absence of biological replication for root chemistry prevents statistical confirmation of preferential Na retention in the belowground tissues. Micronutrient and TE concentrations in roots were generally within the broad ranges reported for vegetable tissues (Kabata-Pendias 2010), although Fe and Cu varied considerably among treatments. No consistent increase was observed across all wastewater treatments, while Cd and Pb were predominantly below their respective analytical quantification limits. Given the descriptive nature of the root dataset, these patterns do not provide evidence of substantial short-term accumulation of potentially toxic metals. Nevertheless, metal-related risks under wastewater reuse may depend on cumulative loading and changes in soil availability over longer application periods (Libutti et al. 2017).

3.3 Limitations and Future Research

The use of standardized synthetic solutions enabled direct comparison of fertigation-water composition and soil effects, but interpretation is limited by the short experiment, two nutrient-poor sandy soils, one crop species, and low replication (two pots per Soil $\times$ Water combination). Consequently, statistical strength was limited, especially for interaction effects, and significant or non-significant results should be considered preliminary evidence rather than definitive or broadly generalizable conclusions. Repeated drainage measurements improved temporal characterization but did not increase biological replication. The synthetic solutions also did not reproduce the variability, microbiological components, or organic complexity of real wastewater. Root elemental composition was based on one analytical result per treatment combination and was interpreted descriptively. Further experiments with greater biological replication are needed to confirm these patterns and estimate treatment effects and Soil $\times$ Water interactions more precisely.

Bulk density was measured before the experiment, whereas porosity and saturated hydraulic conductivity were not determined. Consequently, possible changes in soil structure and hydraulic behaviour could not be directly assessed. Microbial biomass, microbial activity, N-transformation rates and pore-water residence time were also not quantified. Accordingly, the hydraulic, sorptive and microbial processes proposed in the discussion should be regarded as hypotheses consistent with the observed patterns and previous literature rather than as mechanisms demonstrated by the experiment of this study. Drainage volumes were also not recorded for every pot and sampling event, preventing the calculation of cumulative solute loads and complete nutrient mass balances. The leaching assessment was therefore based on concentration patterns and temporal trends. Further research should examine these responses using real wastewater, longer cultivation periods, greater biological replication and field-scale experiments involving a wider range of soils and crops. Future experimental designs should include systematic measurement of drainage volumes, repeated assessment of soil physical and hydraulic properties, and replicated elemental analyses of roots and edible organs. For Na-rich waters, tissue-specific Na distribution should be evaluated together with sensory and market-quality attributes to determine whether tissue accumulation affects the quality and acceptability of the harvested product. These measurements would allow short-term concentration responses to be distinguished from cumulative agronomic and environmental effects.

4 Conclusions

The use of category-specific median compositions synthesized from 75 wastewater-fertigation studies enabled the preparation of chemically defined and reproducible synthetic waters. These solutions captured the central nutrient and salinity characteristics reported for domestic wastewater, industrial effluent and landfill leachate while reducing the influence of extreme observations and avoiding dependence on a single site-specific effluent. The profiles therefore represent standardized, literature-derived compositions rather than universal wastewater characteristics, typical European values or conservative worst-case scenarios. Their use provided a consistent experimental basis for comparing water-composition effects across the soil-plant-drainage system in two nutrient-poor sandy soils with extremely low buffering capacity.

The ionic composition of the fertigation waters was rapidly transmitted to the drainage, particularly in SRSE. Nitrate and, to a lesser extent, ammonium and phosphate frequently exceeded groundwater quality thresholds, especially under IE and L. Sodium, chloride and potassium also showed high mobility, while trace-element concentrations in drainage water remained comparatively low. Significant Soil $\times$ Water $\times$ Time interactions for most drainage variables confirmed that solute transport depended on the specific treatment combination and sampling time. Post-experimental soil chemistry retained significant Water effects for K^+ , Na^+ and Cl^- , indicating that these elements were partly retained in the soil while also being transported through drainage.

Agronomically, nutrient-rich waters improved radish growth relative to the control, resulting in greater aboveground and belowground biomass. Soil and Water significantly affected all biomass variables, whereas their interaction was not significant. The highest total dry biomass was recorded under IE-SRSE. Foliar K, Ca and Mg generally remained within the sufficiency ranges reported for radish, while P remained near the lower sufficiency range. Root elemental composition showed numerical increases in several macronutrients under IE and L, confirming nutrient delivery to the soil-root interface, although the absence of biological replication prevented statistical confirmation of greater nutrient uptake. Na accumulation was pronounced in leaves and numerically high in roots under L and IE, particularly in CSC. Foliar Na reached 25.7 g kg⁻¹ dry matter under L-CSC, identifying a potential crop-quality concern.

These results indicate potential agronomic and environmental risks when nutrient and salt inputs exceed crop demand and soil retention capacity. Although the synthetic waters acted as effective short-term nutrient sources, the observed biomass gains were accompanied by substantial nutrient and salt leaching, particularly under IE and L. Longer-term experiments using real wastewater, systematic drainage-volume measurements, replicated root analyses and field validation are therefore required before wastewater fertigation can be recommended for nutrient-poor sandy soils.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s40710-026-00873-8>.

Authors' Contributions A.O. and L.A. wrote the main manuscript text and produced the figures. M.Z. cured the agronomic part of this study. M.M. and N.C. cured the hydrogeochemical reactions parts of this study, while A.P. cured the treated wastewaters selection and relevance of this study. All authors reviewed the manuscript.

Funding Open access funding provided by Università Politecnica delle Marche within the CRUI-CARE Agreement. This work was partially supported by the University of Zagreb Faculty of Agriculture through the institutional research project Innovative technologies for water resources management for precision irrigation (INOVA), funded under the Recovery and Resilience Mechanism (source 581) within the framework of programmatic funding of public higher education institutions and public scientific institutes in the Republic of Croatia.

Data Availability All data supporting the findings of this study are available within the paper and its Supplementary Information.

Declarations

Competing interests The authors declare no competing interests.

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